CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 4-2024: A 39-Year-Old Man with Fever and Headache after International Travel

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PRESENTATION OF CASE

Dr. Hayden S. Andrews (Medicine): A 39-year-old man was evaluated at this hospital during the summer because of fever with shaking chills, diffuse headache, and fatigue, which had lasted for the 4 days since he had returned from a trip to East Africa.

During the patient's 2-week trip, he had been on a safari in Tanzania but had not had any contact with animals. He reported that he had received mosquito bites, especially during the first days of the trip. He had eaten cooked shrimp and lobster and a small amount of raw tuna. On the second-to-last day of the trip, nausea, vomiting, and nonbloody diarrhea developed. On the last day of the trip, the patient had fatigue and slept for much of the day; however, he felt well enough to travel home the next day.

On the night the patient returned home to Massachusetts, he began to have fever with a temperature of 38.3°C, along with shaking chills, diffuse headache, and lower back pain. The following day, he presented to an urgent care facility because of fever, malaise, and headache. The oral temperature was 38.5°C; the remainder of the examination was normal. Nucleic acid testing was negative for influenza A virus, influenza B virus, and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Oral prednisone was prescribed for the treatment of headache. On the third day home, in the evening, the patient had fever with a temperature of 39.5°C and lower back pain. The next morning, 4 days after returning from the trip, he presented to the emergency department of this hospital for evaluation.

In the emergency department, the patient reported recurrent fevers accompanied by blurry vision, which had occurred despite the use of acetaminophen, ibuprofen, and prednisone. He reported ongoing weakness and anorexia, noting that he was consuming mostly fruits and electrolyte-replacement solutions. He had had nausea with diarrhea but without vomiting. The patient noted that his urine output had decreased and that the urine appeared darker than usual. Review of systems was negative for myalgias, rash, neck stiffness, photophobia, respiratory symptoms, dyspnea, chest pain, abdominal pain, and joint pain. His travel companion reported

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no symptoms. The patient reported no known tick exposures before or during travel in Tanzania.

The patient's medical history was notable for degenerative knee disease with a meniscus tear, for which he had received glucocorticoid injections. There were no known adverse drug reactions. He had received a series of three vaccinations against SARS-CoV-2. He lived with his partner in a suburb of Boston. He had been a competitive athlete and was employed in the energy industry. He drank beer occasionally and did not use tobacco or other substances.

On examination, the temporal temperature was 38.9°C, the pulse 96 beats per minute, the blood pressure 113/62 mm Hg, the respiratory rate 18 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. The weight was 104.3 kg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) was 30.3. Facial flushing and mild scleral icterus were present. The urine was dark brown. The remainder of the physical examination was normal.

The blood levels of calcium, creatine kinase, alkaline phosphatase, total protein, albumin, and globulin were normal; other laboratory test results are shown in Table 1. An extended respiratory viral

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Initial Evaluation
Hemoglobin (g/dl)	13.5–17.5	17.6
Hematocrit (%)	41.0–53.0	51.2
White-cell count (per μl)	4500–11,000	10,060
Differential count (per μl)		
Neutrophils	1800–7700	9850
Lymphocytes	1000–4800	490
Monocytes	200–1200	120
Platelet count (per μl)	150,000–400,000	24,000
Sodium (mmol/liter)	135–145	134
Potassium (mmol/liter)	3.4–5.0	4.8
Chloride (mmol/liter)	98–108	95
Carbon dioxide (mmol/liter)	23–32	25
Urea nitrogen (mg/dl)	8–25	24
Creatinine (mg/dl)	0.60–1.50	1.64
Glucose (mg/dl)	70–110	88
Magnesium (mg/dl)	1.7–2.4	1.1
Alanine aminotransferase (U/liter)	10–55	119
Aspartate aminotransferase (U/liter)	10–40	137
Total bilirubin (mg/dl)	0.0–1.0	3.3
Direct bilirubin (mg/dl)	0.0–0.4	1.5
Haptoglobin (mg/dl)	30–200	<10
Fibrinogen (mg/dl)	150–400	279
Venous blood pH	7.30–7.40	7.41

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

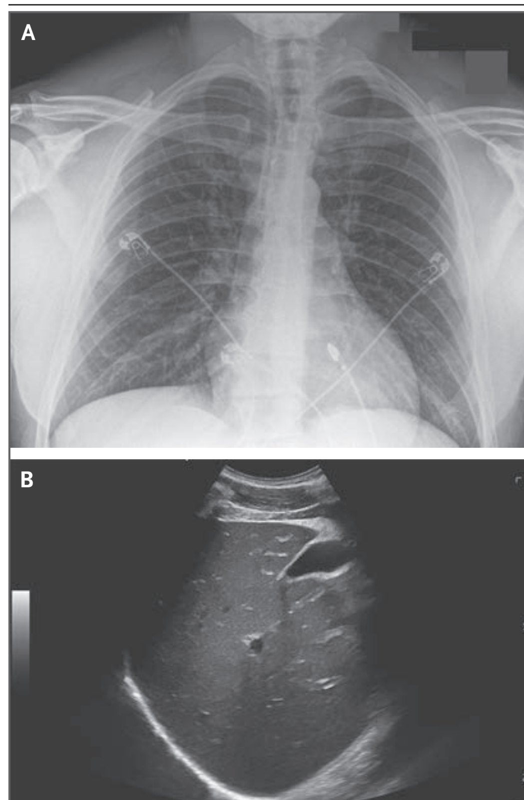

Figure 1. Chest Radiograph and Limited Abdominal Ultrasound Image.

A single-view frontal image from portable chest radiography (Panel A) shows no evidence of pneumonia or pulmonary edema. Electrocardiogram leads are visible. A limited abdominal ultrasound image (Panel B) shows echogenic liver parenchyma indicative of mild hepatic steatosis but is otherwise unremarkable.

panel and a screening test for Lyme disease IgM and IgG antibodies were negative. Urinalysis showed 3+ blood, and microscopic examination of the urinary sediment showed 3 to 5 red cells per high-power field (reference range, 0 to 2). Specimens of blood and urine were obtained for culture. An electrocardiogram showed nonspecific minor T-wave inversions.

Dr. Marc D. Succi: A single-view frontal image from portable chest radiography (Fig. 1A) showed clear lungs without evidence of pneumonia, pulmonary edema, or pleural effusion. The heart size was normal, and there was no radiographic evidence of mediastinal lymphadenopathy. The bones were normal. A limited abdominal ultrasound image (Fig. 1B), obtained to evaluate the liver and biliary system only, showed mild hepatic steatosis but was otherwise unremarkable.

Dr. Andrews: Lactated Ringer's solution and magnesium sulfate were administered intravenously, and acetaminophen and ibuprofen were administered orally. The patient was admitted to the hospital.

Diagnostic and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Edward T. Ryan: I am aware of the final diagnosis in this case; however, much can be garnered by considering the differential diagnosis and clinical approach to this patient's presentation. We will consider the cause of a febrile illness in a previously healthy 39-year-old man who resides in the United States and recently traveled to Tanzania for 2 weeks. Two days before he returned home from this trip, nonbloody diarrhea developed in the patient. He began to have fever, chills, lower back pain, and headache immediately upon returning home. Despite these symptoms and signs, there was an absence of findings on physical examination; he did not have any obvious skin lesions, pulmonary abnormalities, organomegaly, or lymphadenopathy. However, there were several striking laboratory abnormalities, including thrombocytopenia, elevated blood levels of aspartate aminotransferase and alanine aminotransferase, hyperbilirubinemia, mild renal dysfunction, a low haptoglobin level, and hematuria (3+) with only few red cells noted on urinalysis. The white-cell count was normal, and there was evidence of possible hemoconcentration, a finding suggestive of dehydration or capillary leak syndrome.

When evaluating a patient for possible causes of fever after international travel, clinicians consider the destination and the time since the trip, any risk factors, the constellation and duration of symptoms and signs, abnormalities on physical examination and laboratory analysis, and imaging findings, if pertinent.[1] A febrile illness in an international traveler may be related to the trip causally or only temporally. The illness could be caused by an infectious disease endemic in either the destination or home region, a common process (e.g., a urinary tract infection, pneumonia, or influenza), or a noninfectious disease. In constructing a differential diagnosis, I will consider two central clinical features of this patient's case: nonbloody diarrhea and a febrile illness manifesting at the end of a 2-week trip to sub-Saharan Africa.

COMMON CAUSES OF DIARRHEA

Diarrhea is a common ailment reported after international travel,[2] and an acute diarrheal syndrome with fever and myalgias can be caused by a number of invasive enteric pathogens, including salmonella, shigella, campylobacter, and yersinia species. Diarrhea with fever is also associated with a number of viruses, such as enterovirus and norovirus, although the degree and duration of fever in this patient would be atypical for an infection with one of those entities. The patient had clinically significant thrombocytopenia with normal white-cell and differential counts, as well as laboratory data suggestive of hemolysis, including a low haptoglobin level and hemoglobinuria (hematuria [3+] with a paucity of red cells on urinalysis). An intestinal infection with *Shigella dysenteriae* or enterohemorrhagic *Escherichia coli* can be associated with bloody diarrhea, hemolytic anemia, and renal dysfunction; however, *S. dysenteriae* infection is relatively rare, and patients with enterohemorrhagic *E. coli* infection usually have mild or no fever. Moreover, this patient described the diarrhea as nonbloody.

THROMBOTIC THROMBOCYTOPENIC PURPURA

Thrombotic thrombocytopenic purpura should also be considered as a noninfectious cause of fever, headache, renal dysfunction, and hemolytic anemia. In this case, we were not told whether schistocytes were present on examination of the peripheral-blood smear.

VIRAL SYNDROMES

Arboviral infections, a common cause of febrile illnesses associated with international travel, include dengue virus, chikungunya virus, and Zika virus infections, among others. The priority should be to rule out dengue fever, since this illness can be severe and supportive care can substantially improve outcomes. Dengue fever usually develops in patients within a few days after a mosquito bite and can manifest as a nonspecific febrile illness with headache, body aches, leukopenia, thrombocytopenia, and elevated aspartate aminotransferase and alanine aminotransferase levels; dengue shock syndrome may lead to capillary leak syndrome with hemoconcentration. Many patients with dengue fever have a flush of petechiae distal to the tightening of a tourniquet on an arm, a sign of capillary fragility. Hemolytic anemia, however, is not common in patients with dengue fever, and this patient did not have leukopenia.

Other viral syndromes that could explain some of the features present in this patient include Lassa fever, yellow fever, Crimean–Congo hemorrhagic fever, and filovirus infections, such as Ebola virus disease or Marburg virus disease. However, Lassa fever is endemic in West Africa, not East Africa. Tanzania is considered to be in an area at low risk for yellow fever transmission, and the Centers for Disease Control and Prevention (CDC) generally does not recommend yellow fever vaccination for travel to Tanzania. Crimean–Congo hemorrhagic fever is associated with the occurrence of a tick bite and is most often reported in agricultural workers or those working with animals. Cases of Ebola virus disease have not been reported in Tanzania. There was an outbreak of Marburg virus disease in northern Tanzania in 2023, but no cases were reported when this patient visited. We are unaware of any outbreaks of viral hemorrhagic diseases at his destination, and the patient did not report traveling to a region affected by the outbreak of Marburg virus disease or participating in activities associated with filovirus transmission, including contact with infected humans or animals.

BACTERIAL INFECTIONS

A diagnosis of leptospirosis could explain several of the features of this patient's presentation, including the nonspecific febrile illness, elevations in hepatic aminotransferase and bilirubin levels, thrombocytopenia, and renal dysfunction; however, hemolytic anemia would be an atypical feature. Rickettsial infections, including African tick typhus (caused by *Rickettsia africae*), can also cause a nonspecific febrile illness. However, a dermatologic examination did not show abnormalities consistent with a corresponding tick bite, such as an eschar. Typhoid fever is another cause of a nonspecific febrile illness in international travelers. *Salmonella typhi*, which causes typhoid fever, enters the body through the intestinal system; however, diarrhea is not a common feature during the febrile stage, which usually occurs 1 to 3 weeks after the ingestion of organisms. This patient's degree of thrombocytopenia and hemolytic anemia would not be typical features of typhoid fever. Plague is a possible consideration, but this patient did not have any buboes or pulmonary findings that would be consistent with this diagnosis. Tanzania is south of the African meningitis belt, and no petechiae were reported; therefore, the likelihood of invasive meningococcal disease is low.

PARASITIC DISEASES

East African trypanosomiasis can occur in Tanzania, but no tsetse fly bites were reported, and no skin chancres were noted on examination. Babesiosis can cause hemolytic anemia and fever; the patient is a resident of New England, and he presented with a febrile illness during the summer. If babesiosis were the diagnosis, it would be unrelated to his travel, only clinically manifesting toward the end of his 2-week overseas trip. Babesiosis and East African trypanosomiasis can be diagnosed by means of examination of a peripheral-blood smear. These illnesses are not usually associated with diarrhea, but it is possible that the diarrhea and febrile illness are unrelated and represent two distinct processes. However, the diagnosis most likely to cause a febrile illness with hemolysis, hyperbilirubinemia, a normal white-cell count, and thrombocytopenia, presenting at the end of a 2-week trip to East Africa in a person who did not receive chemoprophylaxis, would be malaria, specifically that caused by *Plasmodium falciparum*.

MALARIA

Most cases of malaria diagnosed in the United States occur during the U.S. summer months among patients who have traveled to sub-Saharan

Africa.[3] The majority of cases are diagnosed in patients who did not receive chemoprophylaxis. People born overseas who return there to visit friends or relatives are also at particular risk. Five species of malaria parasites can infect humans, although most travelers infected in Africa acquire *P. falciparum*. Most patients infected with *P. falciparum* present with a nonspecific febrile illness with headache, chills, and myalgias, symptoms that usually develop within 4 to 12 weeks after the infectious mosquito bite.[3] Diarrhea can also occur, as well as cough or shortness of breath. In 2023, cases of malaria caused by *P. vivax* were diagnosed in patients residing in Florida and Texas who had not traveled, which indicated local acquisition.[4] These cases show that mosquitoes capable of transmitting malaria exist in the United States and that limited local or ongoing transmission can occur after the importation of malaria into home communities.

This patient was evaluated for influenza, Covid-19, and tickborne illnesses endemic in New England. The recommendation was to return for further evaluation for malaria if the initial evaluation was negative and fever persisted. The patient was also empirically treated with glucocorticoids for unclear reasons. Caring for a patient with fever after international travel requires a thorough evaluation, which includes obtaining a detailed history of pertinent exposures, epidemiologic factors, immunization status, and chemoprophylaxis status, as well as performing a meticulous physical examination. *P. falciparum* malaria can be rapidly fatal. When malaria is considered in the differential diagnosis, an evaluation for malaria needs to be performed quickly, and the treatment approach usually depends on the laboratory test results. Treatment with glucocorticoids was not indicated in this case.

DR. EDWARD T. RYAN'S DIAGNOSIS

Plasmodium falciparum malaria.

DIAGNOSTIC TESTING

Dr. Erik H. Klontz: At this hospital, the workup for malaria begins with a rapid antigen test. If the test is positive for malaria, microscopic examination of thick and thin blood smears is performed to confirm the infection, identify the species, and quantify the level of parasitemia. A single negative rapid antigen test cannot rule out the diagnosis of malaria; repeating the test every 6 to 12 hours for up to 3 days is recommended if suspicion remains that the patient may have malaria.

The rapid antigen test performed at this hospital has two targets: the *P. falciparum* antigen and the panmalarial antigen, which is shared by all plasmodium species. This patient's rapid antigen test was positive for both the *P. falciparum* antigen, represented by the test 1 (T1) band, and the panmalarial antigen, represented by the test 2 (T2) band (Fig. 2A), findings that indicate infection with *P. falciparum*; however, coinfection with another plasmodium species could contribute to the T2 band, so a mixed infection cannot be ruled out on the basis of this test result.

The intensity of the bands cannot be used to estimate the parasite burden, given that a very high parasitemia level can create a prozone effect that weakens the band intensity.[5] If an estimation of the parasitemia level is required before the results from analysis of the Giemsa-stained thick and thin blood smears are available, an examination of a thin peripheral-blood smear with Wright staining or Wright–Giemsa staining can serve as a quick reference; however, follow-up examination with Giemsa staining, which offers better parasite visualization, should be performed.

In this patient, Giemsa staining of a thin blood smear revealed intraerythrocytic ring forms with an extraordinarily high estimated parasitemia level of 36% (Fig. 2B). Ring forms (or immature trophozoites) are common to all plasmodium species; the ring forms of individual plasmodium species cannot be reliably distinguished from each other or from the ring forms of babesia species. The size of the infected red cells and the morphologic features (or notable absence of such features) of mature forms, including schizonts and gametocytes, can aid in identifying the species. With *P. falciparum*, banana-shaped gametocytes can be present, but mature schizonts are usually absent from the peripheral blood because they become sequestered in postcapillary venules.[6]

Although gametocytes were not seen in this patient's blood smear, many ring forms had a "headphone" appearance that is typical of *P. falciparum* (Fig. 2C). Rare maturing trophozoites producing the golden-brown malaria pigment (hemozoin, from parasite digestion of hemoglobin) were also visible in the blood smear, a finding that ruled out babesia infection (Fig. 2D). The combination of the characteristic trophozoite appearance, malaria pigment production, and

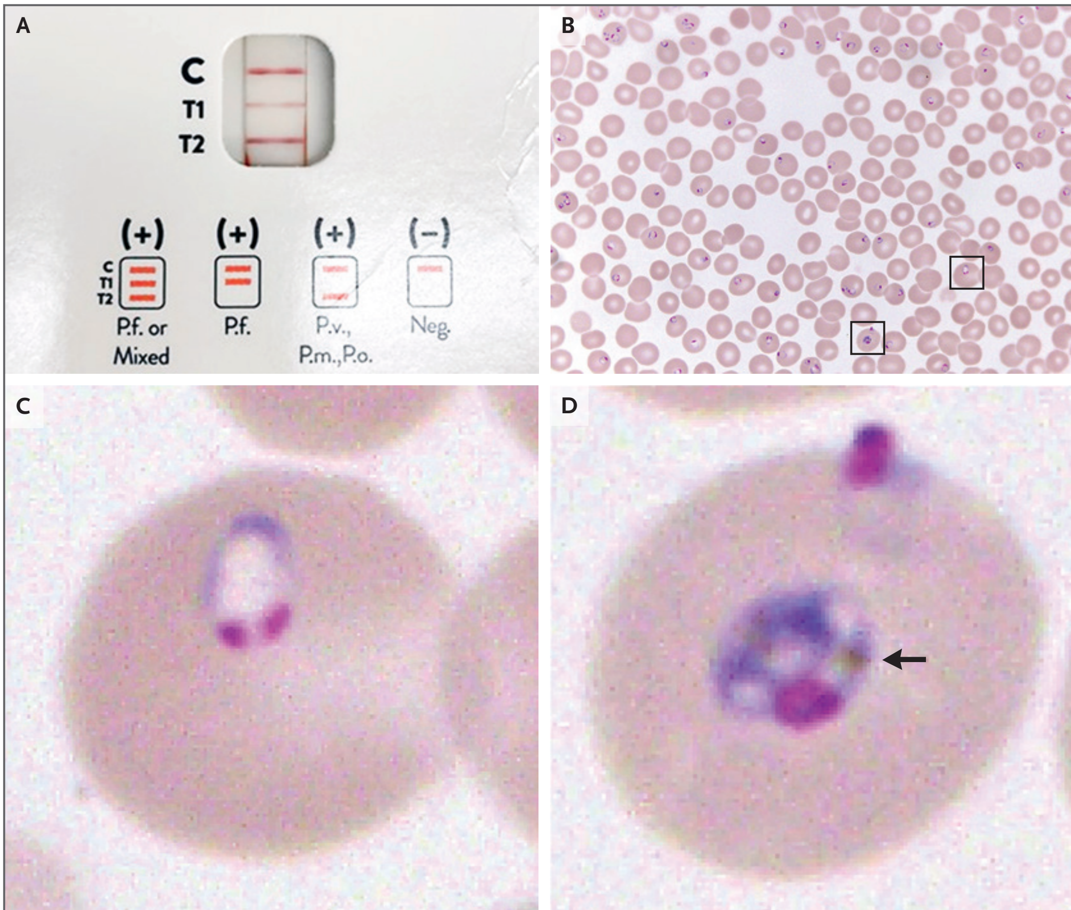

Figure 2. Malaria Test and Blood Smear.

A rapid antigen test for malaria (Panel A) shows bands at the test 1 (T1) and test 2 (T2) lines, which could indicate a *Plasmodium falciparum* infection or a mixed infection including *P. falciparum* and another plasmodium species. Giemsa staining of a thin peripheral-blood smear (Panel B) shows features consistent with *P. falciparum*. There are abundant intraerythrocytic ring forms that are small, thin, and delicate, with an estimated parasitemia level of 36%. Infected red cells are normal in size, and some are multiply infected. Mature forms, including schizonts and gametocytes, are not seen. The right-hand box in Panel B corresponds to Panel C, which shows a ring form with a "headphone" appearance, with two pink dots of chromatin joined by a thin blue band of cytoplasm, a feature characteristic of *P. falciparum*. The left-hand box in Panel B corresponds to Panel D, which shows a maturing trophozoite producing golden-brown malaria pigment (arrow), a feature present in all plasmodium species but absent in babesia species.

the presence of a T1 band on a rapid antigen test confirms the diagnosis of *P. falciparum* malaria. At the same time, the notable absence of mature forms on multiple smears makes coinfection with another plasmodium species extremely unlikely.

MICROBIOLOGIC DIAGNOSIS

Plasmodium falciparum infection.

DISCUSSION OF MANAGEMENT

Dr. Molly L. Paras: Several important pieces of information are needed when selecting treatment for a patient with malaria. The plasmodium species, the antimalarial resistance profile of the parasite according to geographic region, the chemoprophylaxis status of the patient, and clinical features of the patient, including whether the malarial illness can be classified as severe or uncomplicated, should all be considered. The mainstay of treatment for patients with uncomplicated *P. falciparum* malaria usually involves the oral administration of artemisinin-based combination therapy, although other options can be used in specific situations.[7] As compared with other options, the use of artemisinin derivatives is associated with increased survival.[8,9] In the United States, one oral preparation of artemisinin-based combination therapy, artemether–lumefantrine, is commercially available.

Patients who meet the criteria for severe disease are treated with intravenous artesunate.[10] Severe disease can include altered consciousness or coma, severe anemia (defined as a hemoglobin level of <7 g per deciliter), acute kidney failure, acute respiratory distress syndrome, shock, persistent acidosis, jaundice (in the presence of other signs of severe malaria), disseminated intravascular coagulation, and a parasitemia level of at least 5%. Intravenous artesunate has been approved by the Food and Drug Administration, is commercially available in the United States, and is the first-line therapy for severe malaria in the United States. Medical institutions that care for patients at risk for malaria should have intravenous artesunate readily accessible by either stocking it or establishing protocols that facilitate the rapid acquisition of the agent or the rapid transfer of patients to a facility with treatment capacity. It is important to note that if a patient has an infection with either *P. vivax* or *P. ovale*, treatment will need to include a second agent (typically primaquine) that targets the dormant liver-stage form of these plasmodium species, known as the hypnozoite, in addition to including the first agent, which targets the blood-stage form.

This patient received a diagnosis of severe *P. falciparum* malaria on the basis of the presence of acute kidney injury and a parasitemia level of at least 5%. Shortly after an infectious disease consultation was obtained, intravenous artesunate therapy was initiated. At this time, the patient had an estimated parasitemia level of 36%. The parasite burden was monitored daily with Giemsa staining of thick and thin blood smears. After 48 hours of therapy, the parasitemia level decreased to 0.9%, and after 72 hours of therapy, it was less than 0.1%. Once the patient was reliably eating and drinking and the parasitemia level was less than 1.0% at 4 hours after the last dose of intravenous artesunate, his treatment was transitioned to a full course of oral artemether–lumefantrine. His condition remained hemodynamically stable without progressive central nervous system or respiratory symptoms throughout the course of treatment, and his kidney injury had resolved by the time of discharge.

This patient was counseled on the prevention of malaria during future travel. Travelers visiting areas with malaria should be advised on ways to minimize their risk of exposure. Such strategies — implemented during the period from dusk to dawn, when malaria-bearing mosquitoes are most active — include sleeping in rooms with screens or under netting, wearing long-sleeved clothing, and using mosquito repellents on skin and insecticide on clothing and netting. Chemoprophylaxis is also an important tool for preventing malaria in international travelers. Several agents are available, including atovaquone–proguanil, doxycycline, mefloquine, primaquine, tafenoquine, and chloroquine. The choice of chemoprophylactic agent is based on the risk at the destination, the resistance patterns of the parasites at the destination, the medical condition of the traveler, and cost and convenience. Up-to-date, destination-specific recommendations are available from the CDC.[11] Unfortunately, a minority of travelers visiting areas where malaria is endemic seek medical advice before international travel or adhere to malaria chemoprophylaxis. Together with a growing number of travelers visiting destinations where malaria is present, these factors have been associated with an increase in the number of cases of travel-related malaria reported in the United States.

Given the potential risk of postartemisinin delayed hemolysis, in accordance with CDC recommendations, this patient was scheduled to undergo serial laboratory monitoring checks over the several weeks after discharge. His follow-up tests revealed worsening anemia and hemolysis, with thick and thin blood smears negative for malaria; these findings are consistent with delayed hemolysis. The laboratory data were monitored closely, but he did not receive a blood transfusion. It is hypothesized that a delayed clearance of pitted red cells caused by the expulsion of malaria parasites triggers the hemolytic episodes, and patients with a high parasitemia level are at an increased risk for hemolysis.[12]

FINAL DIAGNOSIS

Plasmodium falciparum malaria.

This case was presented at the Medicine Case Conference.
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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